

5.2 Matter

This part of the module considers how kinetic theory explains the behaviour of matter in probabilistic and mechanical terms. The beginnings of the basis of thermodynamic thinking appear in the study of the Boltzmann factor.

The first section (5.2.1) explains ideal gas behaviour in terms of the kinetic theory.

The second section (5.2.2) introduces the Boltzmann factor as the link between energy and temperature. The important idea that differences drive change is introduced here.

5.2.1 Matter: very simple

Learning outcomes	Additional guidance
(a) <i>Describe and explain:</i>	
(i) energy transfer producing a change in temperature and the concept of specific thermal capacity c	HSW1, 2, 8
(ii) the behaviour of ideal gases	
(iii) impulse $F\Delta t = \Delta p$	
(iv) the kinetic theory of ideal gases	assumptions are that the particles (atoms or molecules) occupy negligible volume, that all collisions are perfectly elastic, and that there are negligible forces between particles except during collision HSW2
(v) temperature as proportional to average energy per particle; average energy $= \frac{3}{2} kT \approx kT$ as a useful approximation	
(vi) random walk of molecules in a gas: displacement in N steps related to $\sqrt{N}$.	
(b) <i>Make appropriate use of:</i>	
(i) the terms: ideal gas, root mean square speed, absolute temperature, internal energy, Avogadro constant, Boltzmann constant, gas constant, mole	
<i>by sketching and interpreting:</i>	
(ii) relationships between p , V , N and T for an ideal gas	M3.12
(iii) force–time graph for an interaction with area under line representing the impulse.	

(c) *Make calculations and estimates of:*

(i) temperature and energy change using
 $\Delta E = mc\Delta\theta$

M2.1

HSW5

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(ii) $pV = NkT$ where $N = nN_A$ and $Nk = nR$

number of moles n and Avogadro constant N_A

(iii) $pV = \frac{1}{3}Nm\overline{c^2}$

M2.2

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(d) *Demonstrate and apply knowledge and understanding of the following practical activities (HSW4):*

(i) using an electrical method to find the specific thermal capacity of a metal block or liquid

links to **5.2.1a(i), c(i)**

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(ii) using appropriate apparatus to investigate the gas laws including determining absolute zero

links to **5.2.1a(ii)**

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(iii) using apparatus to investigate the relationship of volume with pressure, measured either by pressure gauge or differential pressure monitor and data logger.

links to **5.2.1b(ii)**

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